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RV COLLEGE OF ENGINEERING®

(An Autonomous Institution Affiliated to VTU)

III Semester B. E. Regular / Supplementary Examinations Feb/Mar -2025

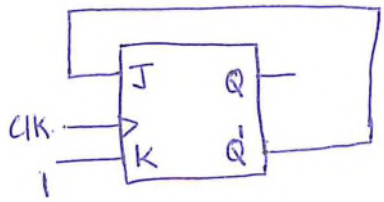
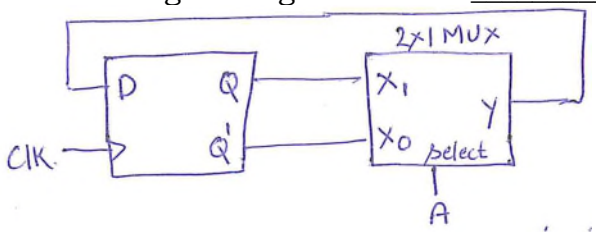
Common to EC / EE / EI / ET

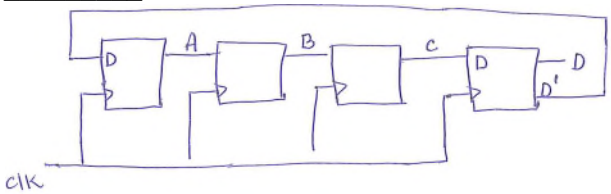
ANALYSIS AND DESIGN OF DIGITAL CIRCUITS WITH HDL

Time: 03 Hours ***Maximum Marks: 100***

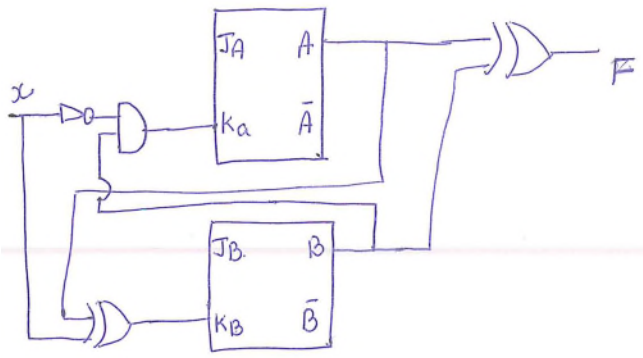
Instructions to candidates:

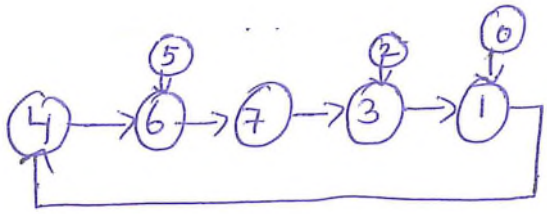
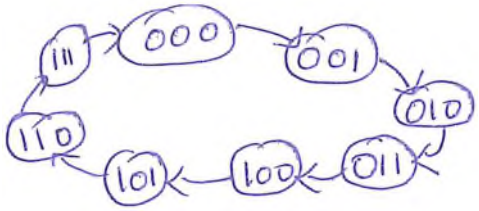
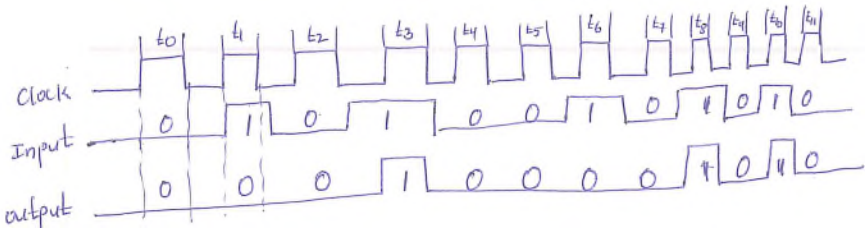
1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.

		PART-A	**M**	**BT**	**CO**
1	1.1	If $A = 6'11 \times 011$, then $^{\wedge}A$ equals _____.	01	1	1
	1.2	The default value for reg data type is _____.	01	1	1
	1.3	For the following Verilog code segment, if the value of A is $12'b110000111001$, what will be the value of $\{A[10:5], 4\{B\}\}$	02	1	2
	1.4	If the 10 bit variable "x" declared as "$reg[9:0]x$" is initialized to $10'b1010101101$, what will be its value after execution of the following code segment? $always\ @(negedge\ clock)$ $begin$ $x <= x << 1;$ $x[0] <= x[9];$ end	02	2	2
	1.5	In FSM, if two transitions from states S_0 and S_1 have the same output for a given input, then it is known as _____.	01	1	1
	1.6	Suppose $reg[0:127]y; reg[0:7]Y_2$ then $Y_2 = y[32-:8]$ is _____.	01	2	2
	1.7	The characteristic equation of JK flip flop is _____.	01	2	1
	1.8	_____ types of case statement will consider both X and Z as don't care.	01	2	1
	1.9	For the given circuit, the initial state Q is 0. The output Q is observed after the each clock pulse. The output sequence at Q is _____.  	02	1	2
	1.10	The state diagram of the given logic circuit is _____.  	02	1	1
	1.11	In a 4 bit ring counter, what is the initial state if the counter starts with 1000?	01	2	1

1.12	If the following counter is initially at 0000 state then after 6th clock. $ABCD$ will be _____.  	02	2	2
1.13	The group of bits 11001 is serially shifted (right-most bit first) into a 5 bit parallel. Output shift register with an initial state 01110. After three clock pulses, the register contains _____.	01	1	1
1.14	For the following Verilog module the output f equal to _____. $module\ Q_1(f, a, b, c);$ $input\ a, b, c;$ $output\ f;$ $wire\ t_1, t_2;$ $nand\ G_1\ (t_1, a, b);$ $nor\ G_2\ (t_2, b, c);$ $or\ G_3\ (f, t_1, t_2);$ $end\ module.$	02	1	2

PART-B

2	a	Explain $FPGA$ design flow with a neat diagram.	04	2	2
	b	Design and write a Verilog code (dataflow) for full adder using mux.	04	2	3
	c	Design a 4-bit Binary Adder and subtractor. Write a Verilog code for the same using structural modeling. Develop a test bench for the same.	08	2	2
3	a	Design a 4-bit comparator using 2-bit comparator. Write a Verilog code using structural modeling. Develop a test bench for the same.	08	2	3
	b	Design a 2×2 array Multiplier with two 2-bit input, "a" and "b" and a $4 - bit$ output, "p" using basic gates. Write a Verilog code using structural level modeling for the same.	08	2	3
		OR			
4	a	Design a priority encode using case statement in order of $6, 4, 2, 0, 5, 3, 1, 7$.	04	2	3
	b	Design a $3{:}8$ decoder using $2{:}4$ decoder and write a Verilog code using structural modeling.	06	2	2
	c	Design and develop a 4-bit carry look ahead adder and write Verilog code using dataflow modeling.	06	2	2
5	a	Construct a transition table and state diagram for the circuit shown below.  	08	3	3

	b	Design a sequential circuit using JK flipflop to satisfy the following equations. $A(t+1) = \bar{A}\,\bar{B}CD + \bar{A}\,\bar{B}C + ACD + A\bar{C}\bar{D}$ $B(t+1) = \bar{A}C + C\bar{D} + \bar{A}B\bar{C}$ $C(t+1) = B$ $D(t+1) = \bar{D}$	08	3	4
		OR			
6	a	Design a synchronous counter for the following count sequence $4-6-7-3-1-4-6$ ... [Avoid lock out condition].  	08	3	3
	b	Design a 3-bit up counter using JK flipflop.  	08	2	3
7	a	Design a 4-bit universal shift register for the following conditions. S_0 S_1 Operation 0 0 No Change 0 1 Shift right 1 0 Shift left 1 1 Parallel load	08	3	4
	b	Design a ripple decade counter using $JK-FFs$ without using $PRESET$ and $CLEAR$ signals with the help of a state diagram and explain. In the same design, show how $MOD-2$ and $MOD-5$ counters can be built to support the design of $IC-7490$.	08	3	3
		OR			
8	a	Construct a 4-bit ring counter using $D-FFs$. Draw the timing diagram for the same. Write the Verilog code for the same.	06	2	2
	b	For the given timing diagram as shown in figure, state the sequence which is detected and develop a Verilog code for mealy FSM, assuming that overlapping of pattern is allowed.  	06	3	3
	c	Design MOD-5 asynchronous counter. (Assume $CLEAR$ input is available in the IC).	04	2	2
9	a	Design an arithmetic circuit with one selection variables and two data inputs A and B. When $S=0$ the circuit has to perform the addition operation $F=A+B$. When $S=1$ the circuit has to perform the increment operation $F=A+1$. Draw relevant arithmetic circuits for 4-bits.	08	3	4

	b	Explain the operation of 4-bit shifter circuit.	08	2	2
		OR			
10	a	Design an arithmetic and logic unit with three selection variables $S_2\ S_1\ S_0$ that generate the following operations. Draw the logic diagram of one typical stage			

Arithmetic operations					Logic operations			
S_2	S_1	S_0	$C_{in} = 0$	$C_{in} = 1$	S_2	S_1	S_0	$C_{in} = X$
0	0	0	$F = A$	$F = A + 1$	1	0	0	OR
0	0	1	$F = A + B$	$F = A + B + 1$	1	0	1	XOR
0	1	0	$F = A - B - 1$	$F = A - B$	1	1	0	AND
0	1	1	$F = A - 1$	$F = A$	1	1	1	$Complement$

			08	3	4
	b	What is scratch pad memory? Analyse the execution sequence in a processor unit with 2-port memory. Draw relevant block diagram.	08	3	3